WAFFLESTUDIO, FALL 2026

Product Engineering

Ship to testers

Session 04, 2026/11/7
Chaehyun Park

LAST TIME

From the design to working code

Divide the work → **Pick the first slice** → **Build it with an agent** → **Test it**

The rest of the MVP is built the same way, one slice at a time.

Today

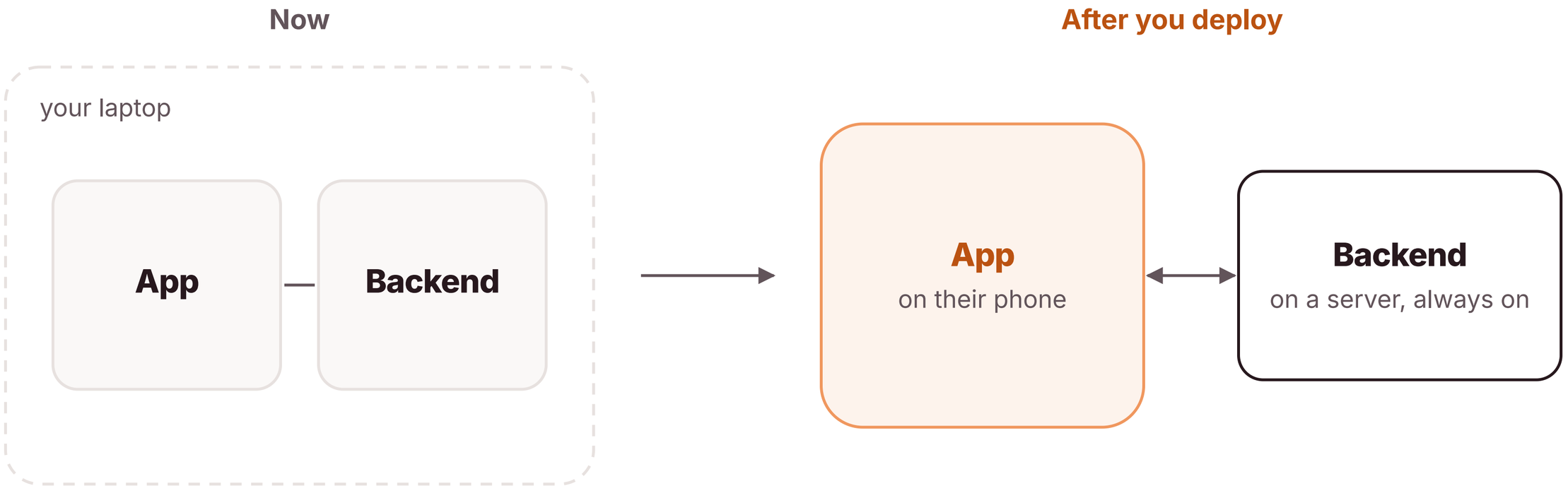
Deploy → Control access → Design the UI → Test with people

01

Deploy

Getting the app onto other people’s phones

What deploying means



Keep two backends

Development

test data

used while building new slices

Production

real users’ data

used by testers’ apps

Some teams add a third, staging, to check changes before they reach production.

Three ways onto a phone

1 · this week

Hand them your phone

No setup

2

Send an install link

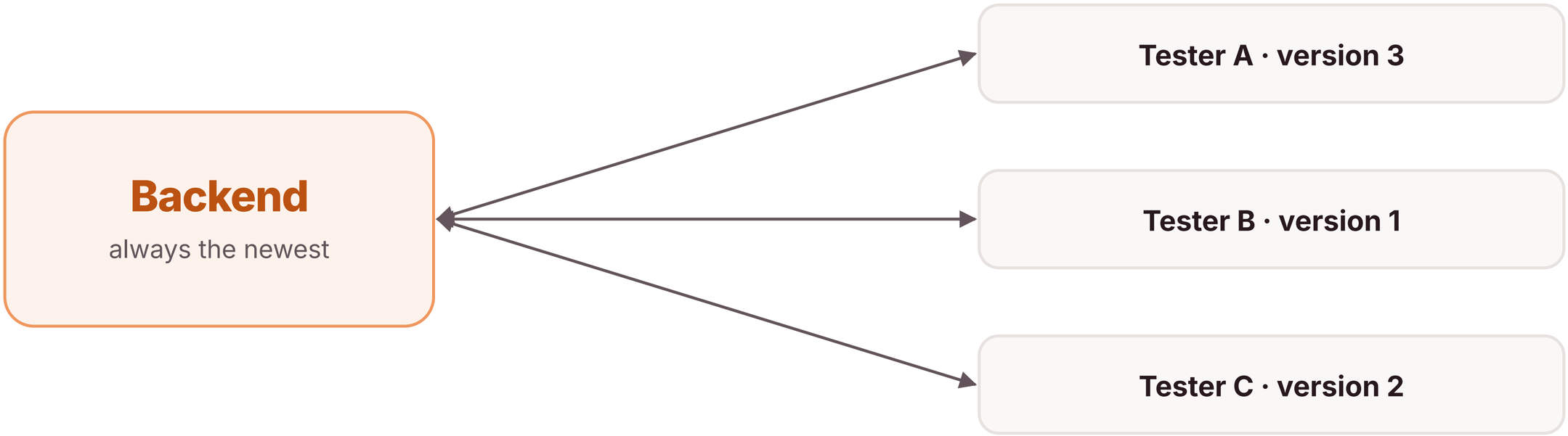
Android: an install link. iPhone: TestFlight ($99/year Apple account).

3 · next week

Release to the store

Needs store review, a store listing and a privacy policy.

Old versions stay on phones



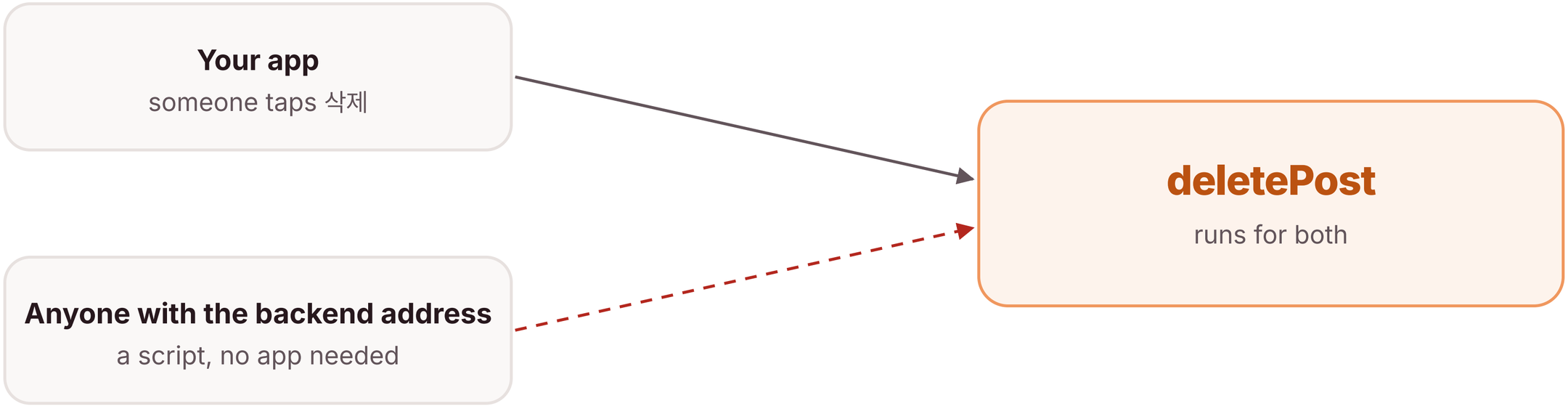
When you change the database, add new fields instead of renaming old ones. Older versions still read the old names.

02

Control access

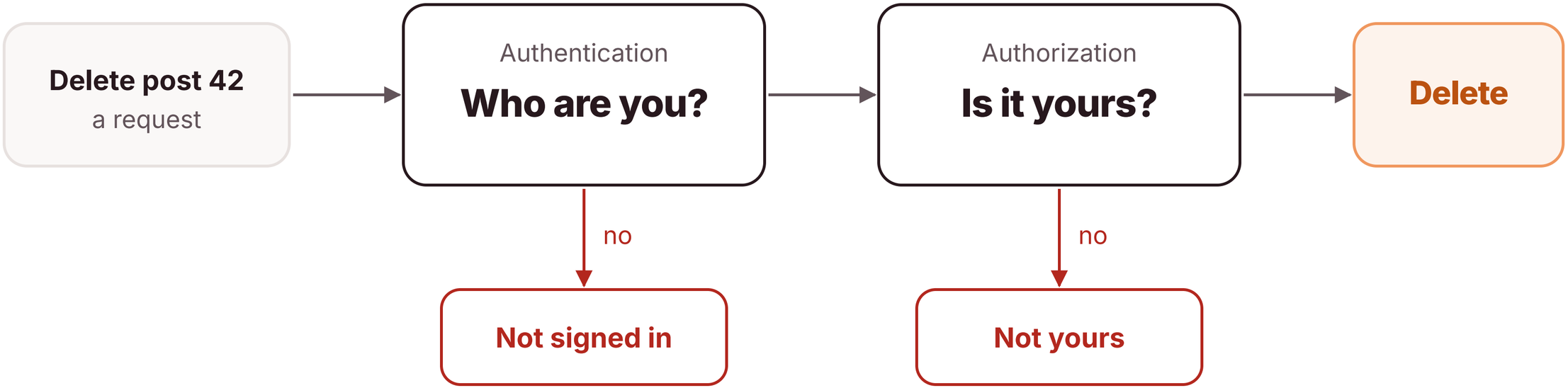
Authentication and authorization

Anyone can call your backend



The backend address is inside your app, so anyone can find it and call your functions directly.

Two checks before changing data



Login handles the first. You write the second yourself, in every function.

OWASP Top 10 — A01 Broken Access Control
top10.owasp.org/2021/A01_2021-Broken_Access_Control

Ask the agent to show you both checks

```
deletePost(postId)
  user = who is asking
  if there is no user          → stop: not signed in
  post = find postId
  if post.author is not user   → stop: not yours
  delete post
```

PROMPT

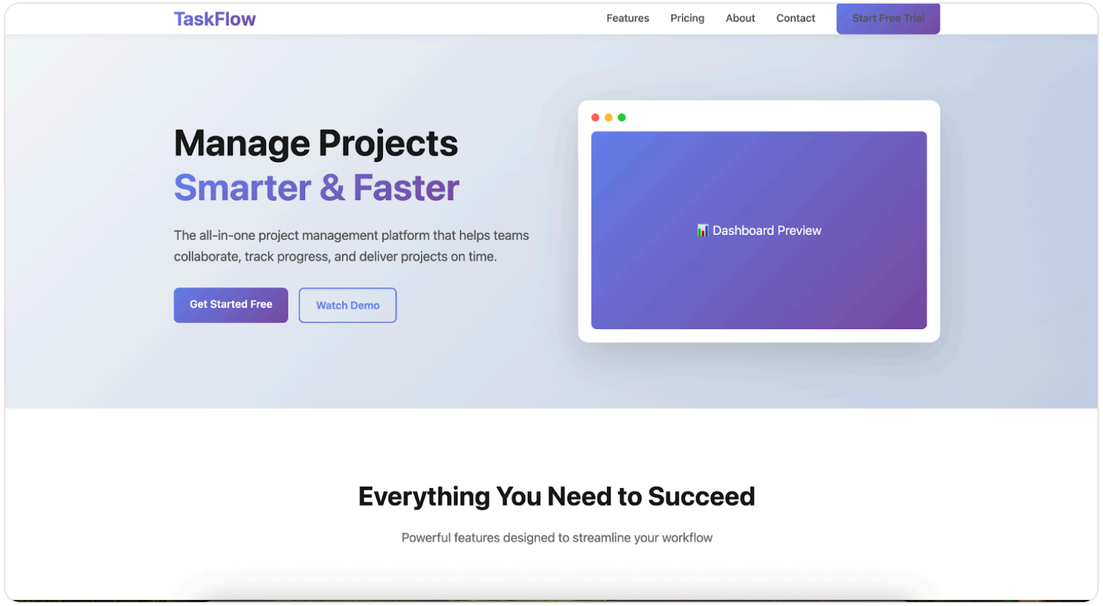
List every backend function that changes data. For each one, show me the line that checks who is asking, and the line that checks it’s theirs.

03

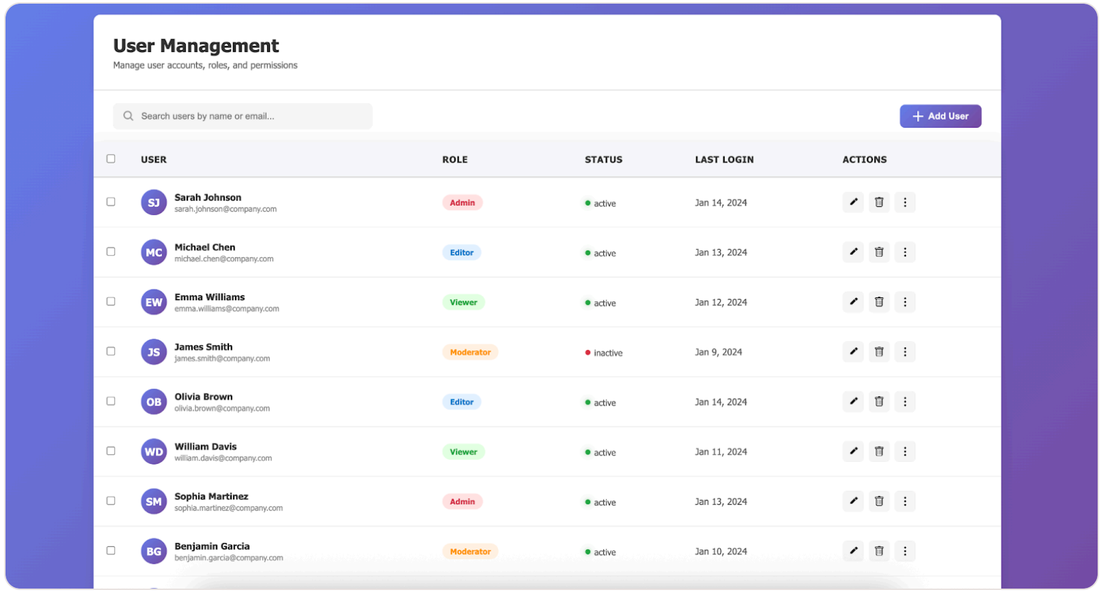
Design the UI

Common mistakes, and what good looks like

What AI makes with no design direction



Landing page

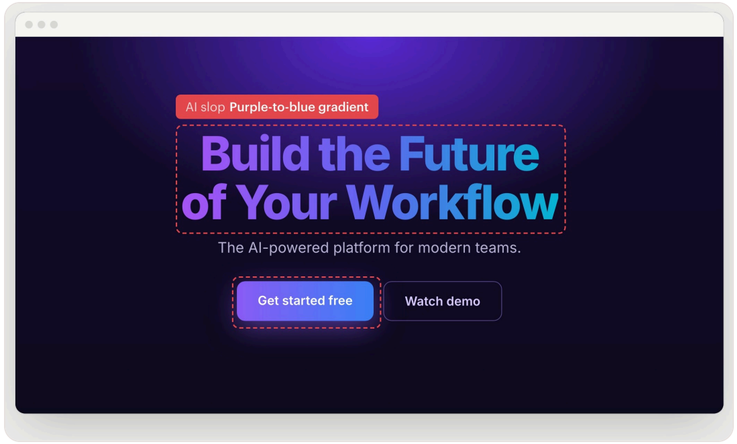


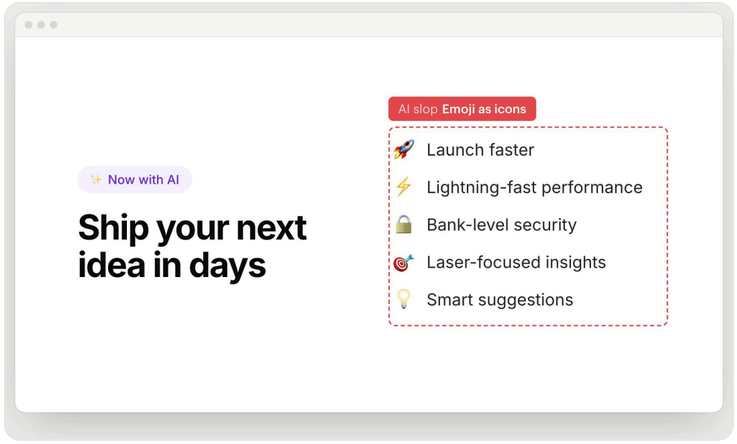
Admin dashboard

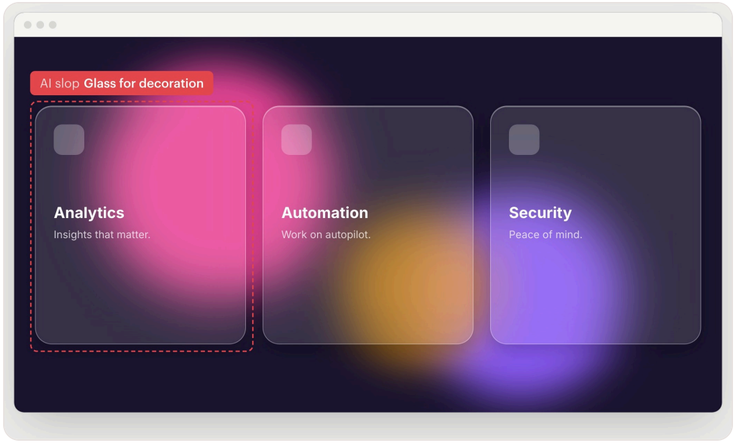
Both made from a one-line prompt with no design instructions.

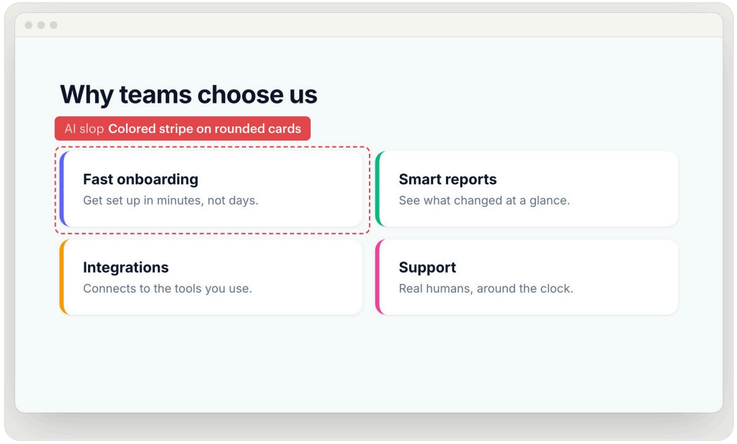
Anthropic — Improving frontend design through Skills (2025/11)
`claude.com/resources/articles/improving-frontend-design-through-skills`

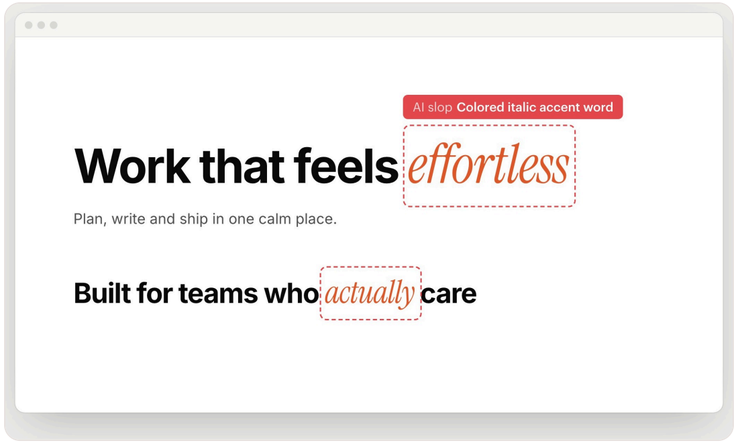
Common AI design patterns

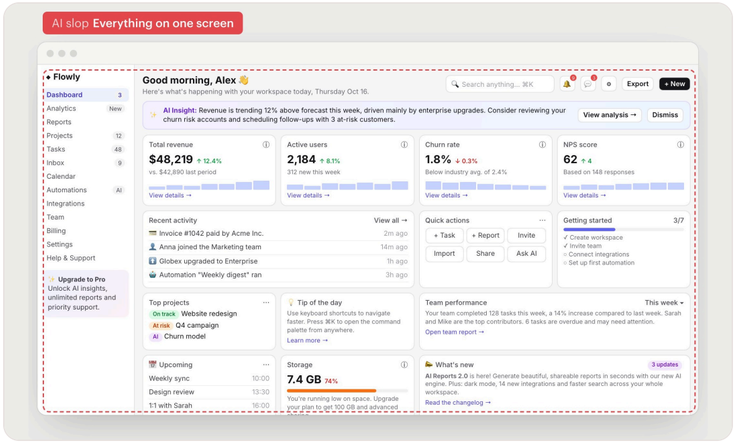




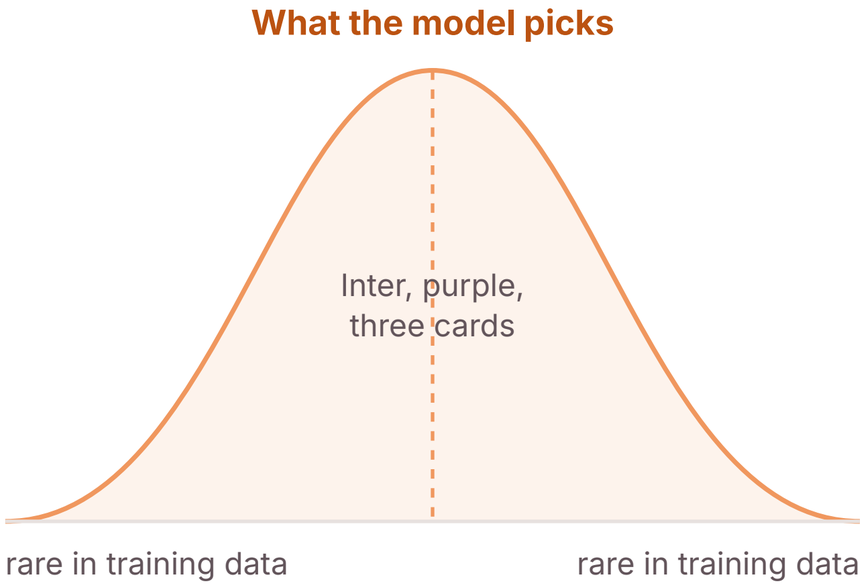








Why it looks the same



Same layout

Centered hero, two buttons, a grid of feature cards

Same look

Gradients, glows, big rounded corners, safe fonts

Same words

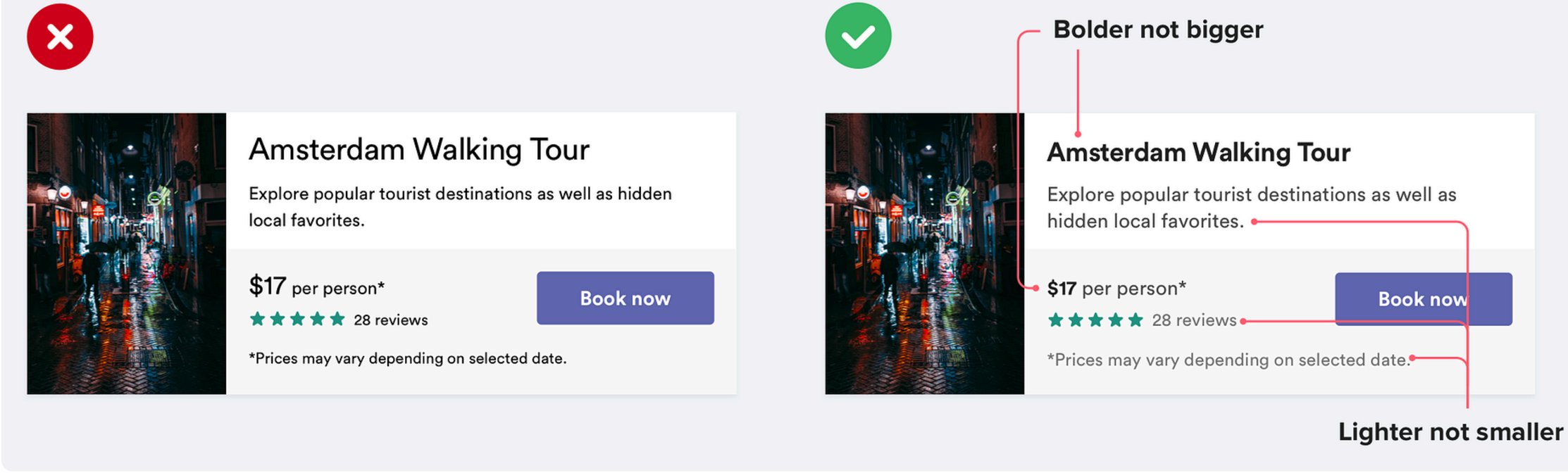
“Everything You Need to Succeed”, “Manage Projects Smarter & Faster”

With no direction, the model picks the most common choice in its training data.

Anthropic (2025/11) · Nick Babich, How to spot AI-generated design (2026/05)
x.com/101babich/status/2057097456857330089

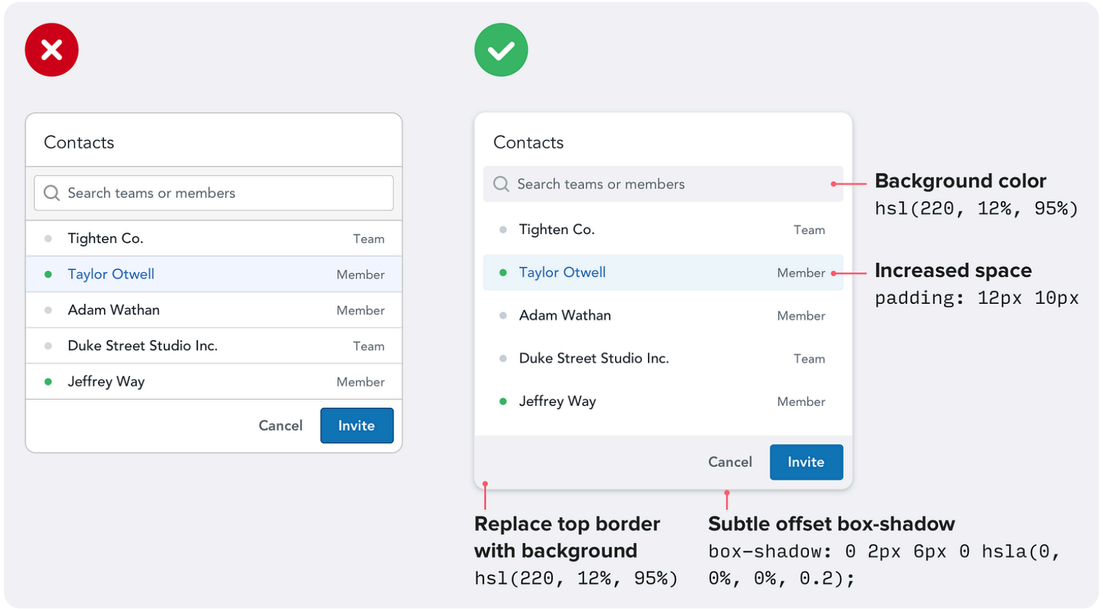
WHAT GOOD LOOKS LIKE

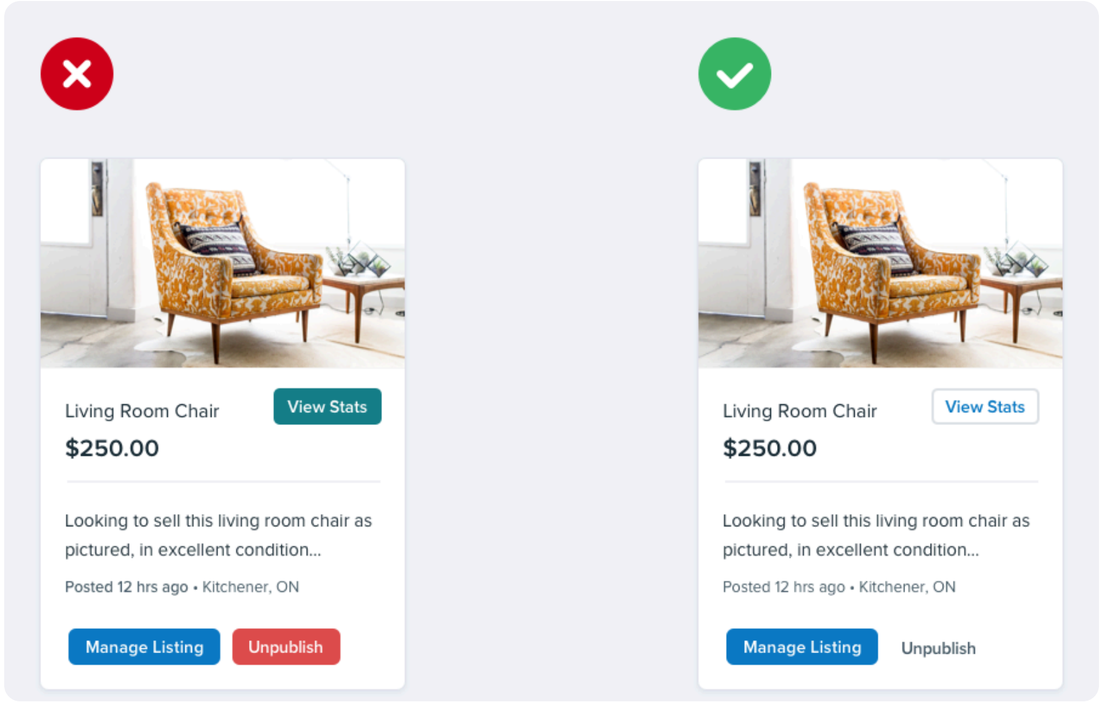
Hierarchy from weight and color, not size



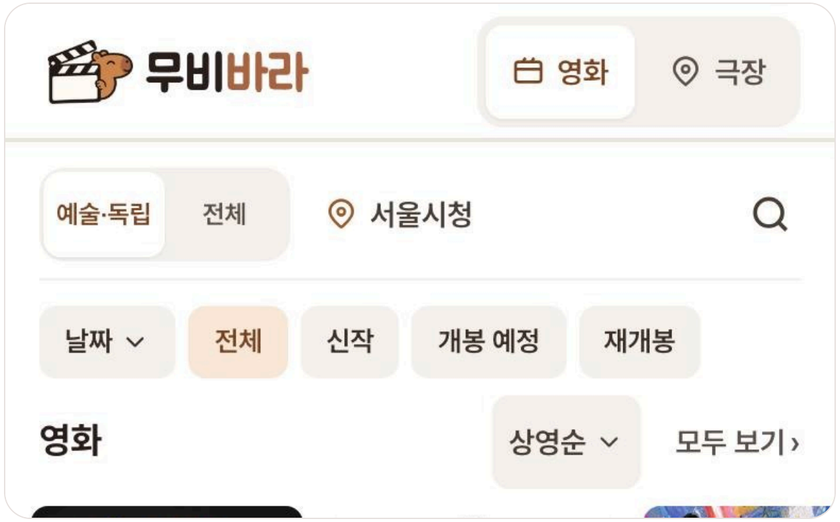
WHAT GOOD LOOKS LIKE

Fewer borders, fewer colored buttons

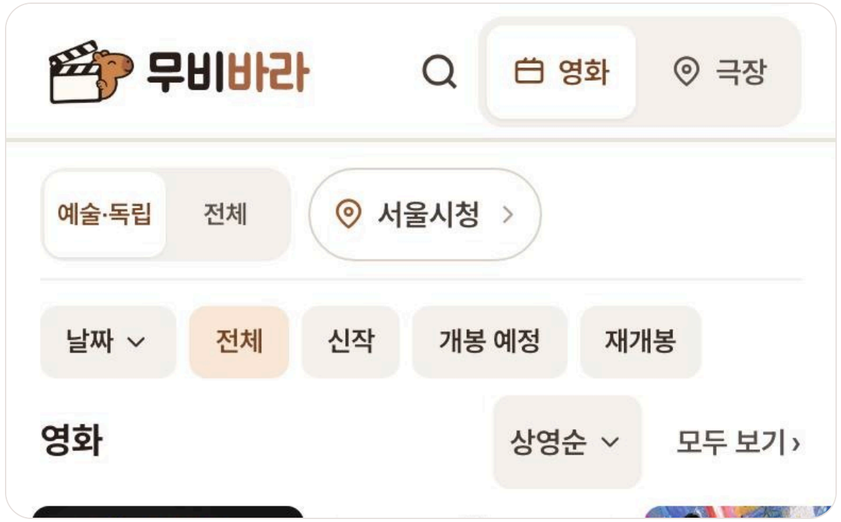




Keep related controls together



Before



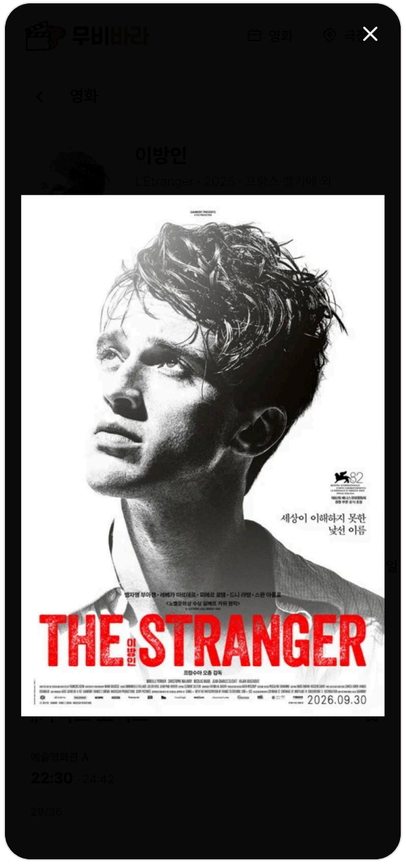
After

The search icon sat next to 서울시청, so people read it as part of the location.

Respond to every tap



Before



After

Someone tapped the poster again and again. Nothing happened, so it looked broken.

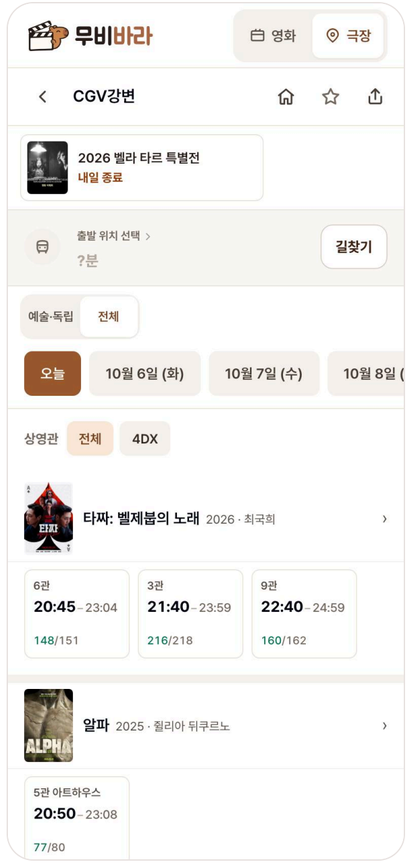
Now it opens full screen.

Nielsen Norman Group — 10 Usability Heuristics, #1 Visibility of system status
nngroup.com/articles/ten-usability-heuristics

Choose defaults carefully



Before

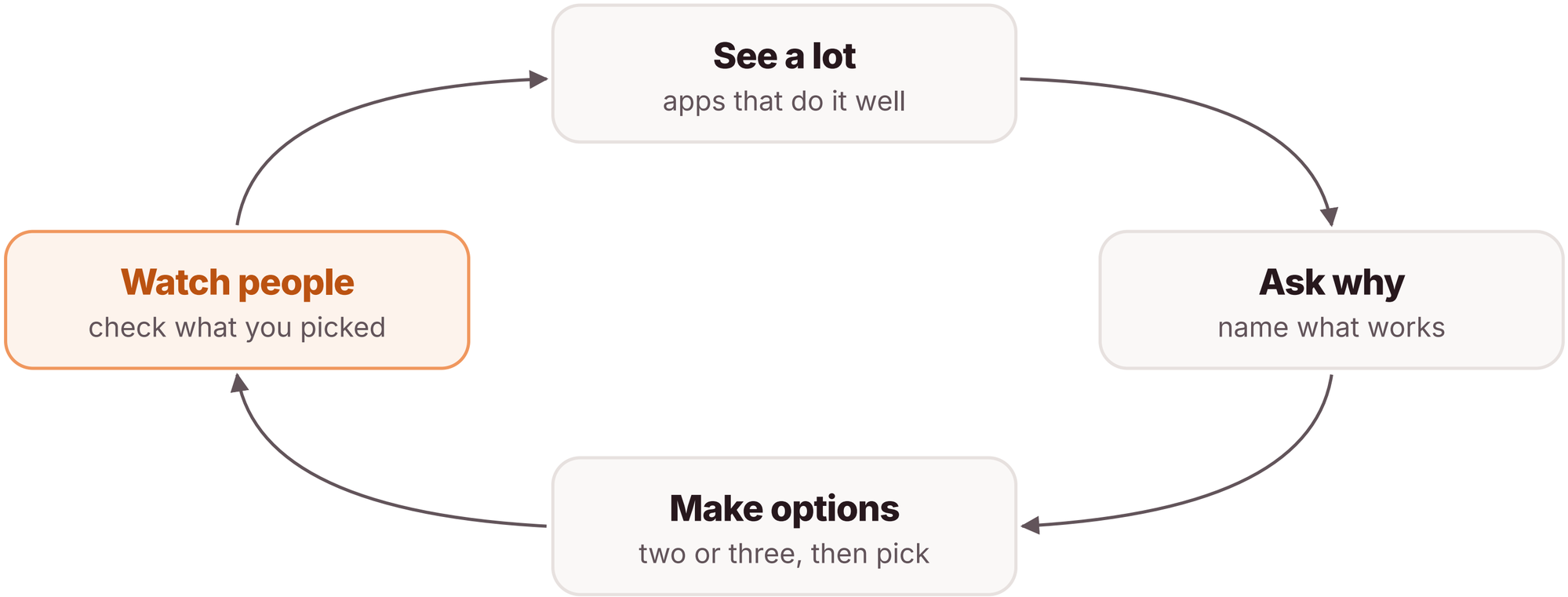


After

예술·독립 was on by default. People saw two films, thought that was all, and left.

A first visit now opens on 전체.

How to build taste



SEE A LOT

Where to look

Real apps, screen by screen	`mobbin.com`
Apple’s design guidelines	`developer.apple.com/design`
Usability principles	`nngroup.com/articles/ten-usability-heuristics`
Visual design tips	`medium.com/refactoring-ui`

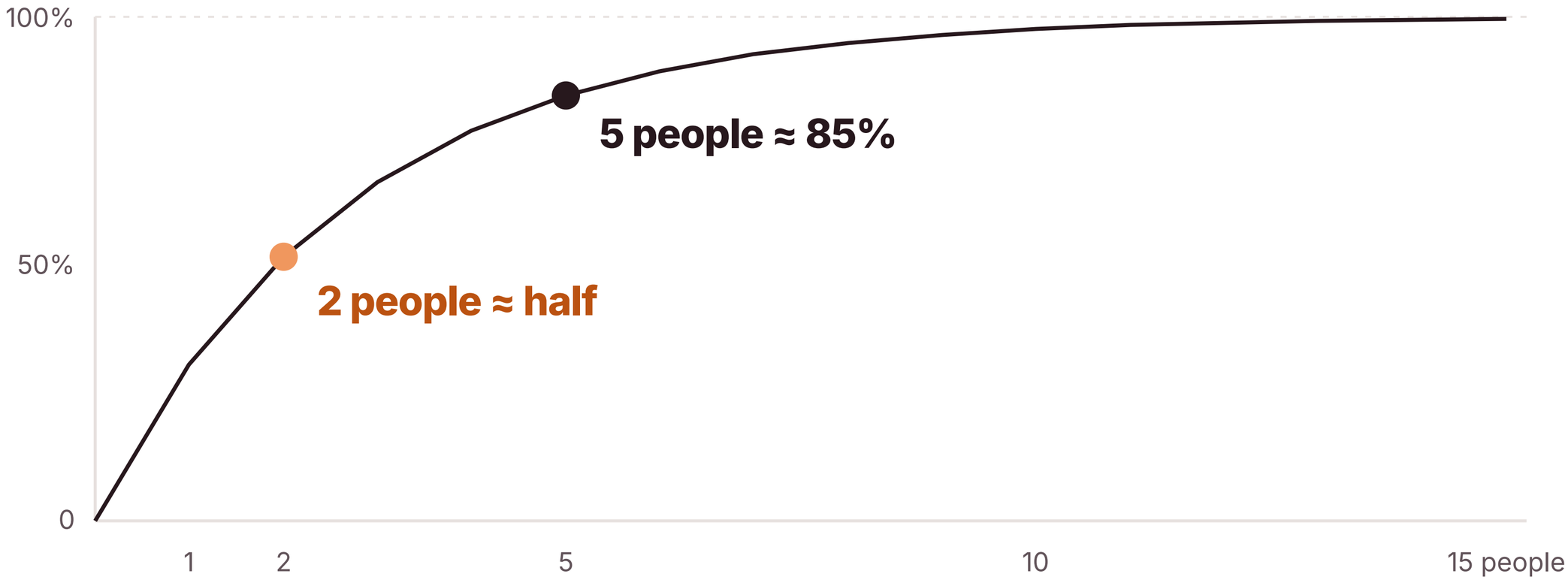
Find three apps that solve your core screen. Write one line on what each does better than yours.

04

Test with people

Usability testing

A few people find most of the problems



Jakob Nielsen — Why You Only Need to Test with 5 Users
nngroup.com/articles/why-you-only-need-to-test-with-5-users

How to run one test

One task → **Think aloud** → **Don’t help** → **Note where they stop**

A task	“Find a film you could see tonight near here.”
Not a task	“What do you think of my app?”

Steve Krug — Usability test script

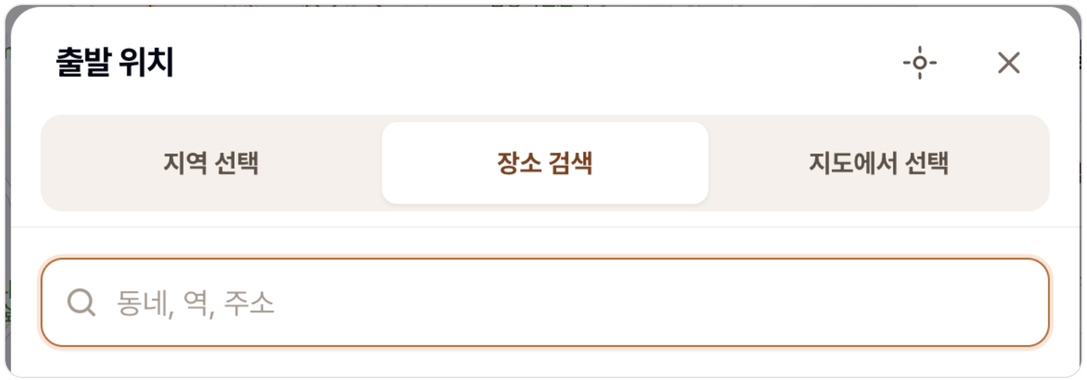
sensible.com/downloads/test-script-web.pdf

Tapping the travel time

Before



tap 대중교통 17분 ↓



It opened 출발 위치. The two map logos had no labels.

After



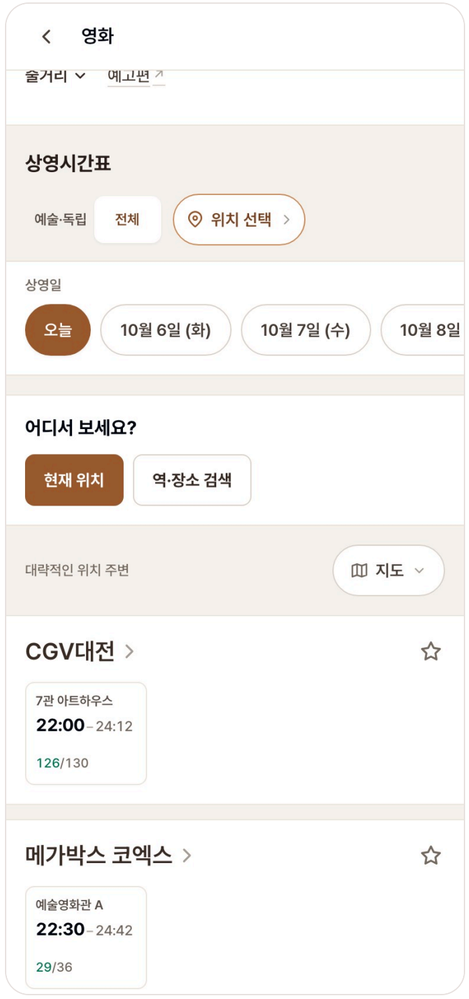
tap 대중교통 17분 or 길찾기 ↓



It opens the route: when to leave, and labelled map apps.

무비바라 FIELD NOTES 03

Write down what you saw



Before · scrolled 790px



After · top of the page

OBSERVATION	In PostHog, 71 visitors had opened film pages and there were 5 taps on the location controls. On a 390px screen the first showtime sat 1,336px down the page, below the cast. > "It's asking for location without telling users what it's being used for, isn't that why people aren't clicking on it?"
INFERENCE	The prompt doesn't say what a location gives you, and visitors arriving from search have to scroll past the credits to reach the times.
PROPOSAL	Answer the search arrival first: put the times under the title and show what a location adds before asking for it. The data was too thin to prove the cause, so this was a bet, not a proof.
CHANGE	`e4f26ea`: 상영시간표 follows the title (its heading moved from 752px to 442px). One bar offers "가까운 순으로 보기 ›", and each cinema shows a grey "🚌 ?분" where its trip time will go. 작품 정보 ↓ jumps to the credits. 기획전 and GVs share one row. Related: `cfd83aa`, `d77dfd7`, `c691fa7`. The first-visit location modal was dropped too (`230799e`). Live
OUTCOME	A re-check of location taps per film-page visitor is due. Not done yet.

2026/10/02

Each week until demo day



05

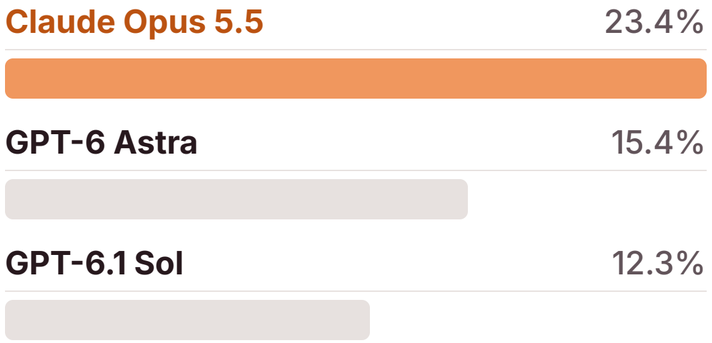
Work with AI agents

Which model, and how I use them

Use Claude Opus 5.5

Coding agents

Arena: people compare two agents on the same task



arena.ai, share of wins, 2026/11/07

Web frontend

Arena WebDev: people pick the better web app

Claude Opus 5.5	1,814
GPT-6 Astra	1,770
GPT-6.1 Sol	1,742

arena.ai, Elo, 2026/11/07

UI design

DesignArena: web, React Native, Android, games

Claude Opus 5.5	1,450
GPT-6.1 Sol	1,422
GPT-6 Astra	1,393

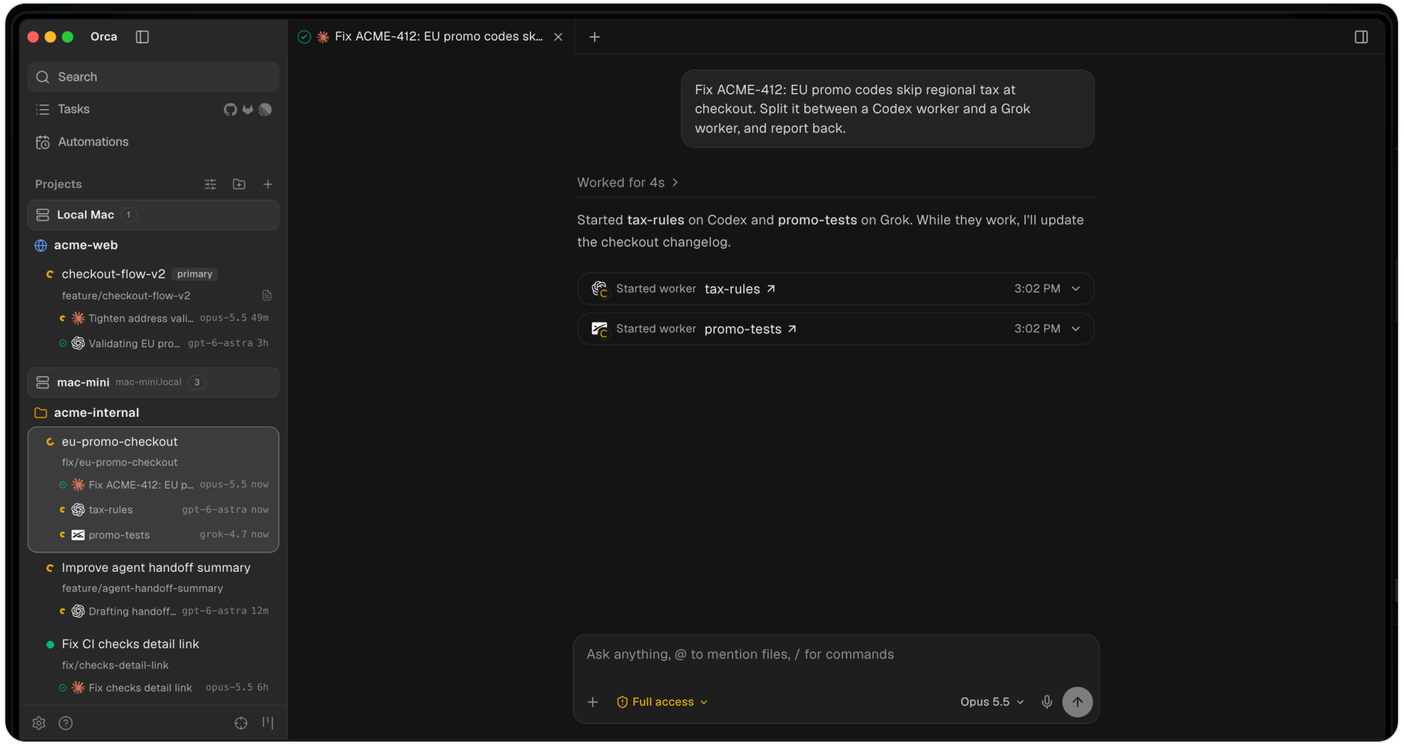
designarena.ai, Elo, 2026/11/07

Codex runs OpenAI's GPT models. Claude Code runs Opus 5.5. All three rankings come from people comparing results without seeing the model name.

Arena — Coding Agents and WebDev leaderboards · DesignArena — Overall Agentic leaderboard

`arena.ai/leaderboard · designarena.ai/leaderboard`

One agent per task, each in its own copy of the repo



A worktree per agent

Agents work in parallel without touching each other’s files.

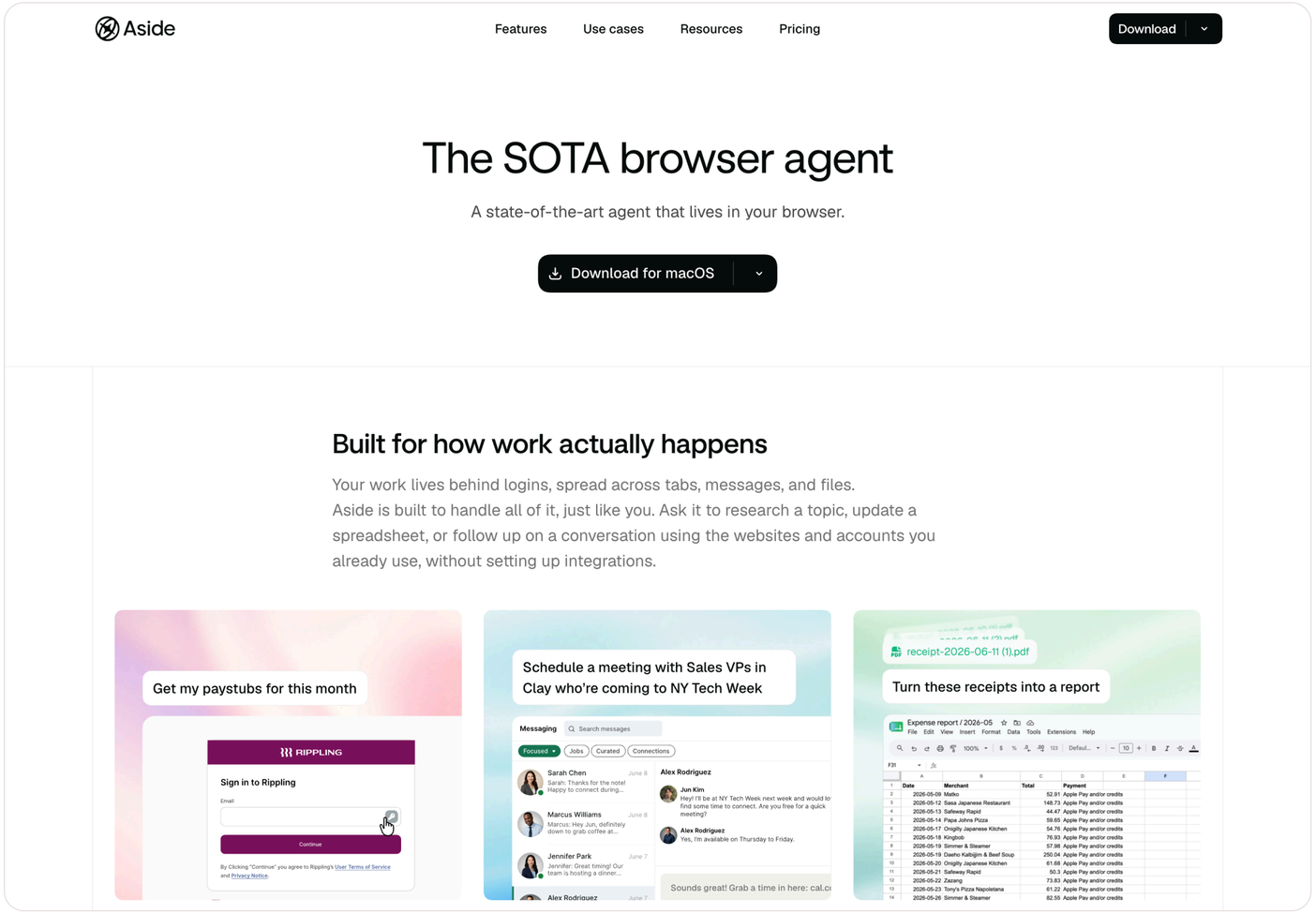
Review the diff, then merge

Each task ends as a branch I read before it lands.

무비바라 so far

159 agent branches across the site, crawler and app.

An agent that uses the browser



Research

The references in this deck were found and checked with it.

Check the real site

The agent opens 무비바라, clicks through a flow and reports what it saw.

Logged-in work

Dashboards and consoles the coding agent can’t reach.

Aside — browser agent

aside.com

06

Assignment 4

Keep building, and test with two people · due 11/13, 23:59

This week

1. **Keep building** the next slices of your core flow. One PR per slice, CI green.
2. **Add the two checks** (signed in? is it theirs?) to every backend function that changes data.
3. **Two people try it,** on your phone or theirs. Write down what you saw.
4. **Fix the one thing** that stopped them most.

What to hand in

1. **Your PRs** for this week's slices
2. **One file,** `docs/04-notes.md`
   Who tried it, what you saw, what you fixed, before and after screenshots.

Due 11/13, 23:59.

On our stack

A production backend	`docs.convex.dev/production`
Access checks in functions	`docs.convex.dev/auth/functions-auth`
An install link for testers	`docs.expo.dev/build/internal-distribution`
Updates without a new build	`docs.expo.dev/eas-update/introduction`

Classes and due dates

9/12	**01**	**Problem**
9/22		**Pitch & sketch** due 23:59
9/29	**02**	**Design & setup**
10/30		**Design & setup** due 23:59
10/31	**03**	**Build end to end**
11/6		**End to end** due 23:59
11/7	**04**	**Ship to testers** today
11/13		**MVP** due 23:59
11/14	**05**	**Run it**
11/20		**Release** due 23:59
11/21	**06**	**Demo day**

Reading

Usability testing on 10 cents a day, Steve Krug	How to run a test
Developing Taste, Emil Kowalski	Building design taste
7 Practical Tips for Cheating at Design, Refactoring UI	Visual design tips
10 Usability Heuristics, Jakob Nielsen	Usability principles

NEXT

05
Run it

11/14 13:00